The application of a deep learning method for kidney stone detection by CT scans: A YOLOv8 approach

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***Abstract* — Kidney stone is a common urological condition that causes a significant healthcare burden. Early detection is crucial for optimal treatment planning. Traditionally, urologists rely on manual analysis of Computed Tomography (CT) scans for kidney stone detection which is a time-consuming and potentially error-prone process. This study investigated the application of a deep learning approach, specifically YOLOv8, for automated kidney stone detection in CT scans. We evaluated the performance of YOLOv8, with the YOLOv8x model demonstrating exceptional performance, achieving a near-perfect F1 score of 0.99 in detecting kidney stones on CT scans. This approach improves diagnostic accuracy by providing radiologists and urologists with a computer-aided diagnosis (CAD) tool for faster and more accurate kidney stone detection by CT scans**.

Keywords — Kidney stone; Deep learning, YOLO, Object detection, Medical imaging, Model evaluation

I. Introduction

Kidney stone is a global health issue, with a prevalence of up to 11% in adults in the USA (1). In Thailand, however, the rate is even higher reaching up to 16.9% (2). The kidney stones within the urinary tract give rise to intricate complications and significant morbidity among diverse patient populations, manifesting as symptoms such as pain, urinary tract infections, and the potential for acute or chronic renal failure (3, 4). Particularly, the one located within the renal pelvis can induce severe pain, obstruct the urinary pathway, and lead to renal failure.

Diagnostic imaging plays a crucial role in identifying and guiding the treatment of kidney stones. However, recommendations vary among institutions regarding the initial imaging method for suspected obstructive nephrolithiasis. Despite these differences, non-contrast CT of the abdomen and pelvis consistently provides the most accurate diagnosis (5, 6).

While detecting kidney stones in CT scans is generally considered achievable, the process can be time-consuming and susceptible to errors, especially under pressure and high workloads. This is particularly relevant in the current healthcare landscape where CT scan requests are on the rise.

In the domain of medical image analysis, accurate and efficient kidney stone detection in CT scans plays a critical role in facilitating early diagnosis and treatment. However, traditional manual analysis methods are often time-consuming and prone to human error.

Recent advancements in deep learning, a subfield of artificial intelligence, have revolutionized the field of medical image analysis. Deep learning algorithms offer the potential to automate tasks traditionally performed by radiologists, including the detection of kidney stones in CT scans. This automation has the potential to improve diagnostic efficiency, accuracy, and consistency, ultimately leading to better patient care.

One promising deep learning architecture for automated kidney stone detection is You Only Look Once (YOLO). YOLO algorithms are known for their speed and accuracy in object detection tasks (7). This study investigates the application of YOLOv8, the recent iteration of the YOLO architecture, for automated kidney stone detection in CT scans. We evaluate the performance of YOLOv8 in terms of accuracy, precision, recall, and other relevant metrics to assess its potential as a computer-aided diagnosis (CAD) tool for radiologists.

II. Materials and methods

A. Patient datasets

This study was a retrospective diagnostic analytic study, which included kidney stone patients who were treated at the Srinagarind Hospital, Faculty of Medicine, Khon Kaen University, Thailand between January 1, 2017, and December 31, 2021. The inclusion criteria were 1) 18 years old or older, 2) confirmed diagnosis of kidney stones after stone removal, and 3) available CT images in Picture Archiving and Communication System (PACS). Patients were excluded if the images contained serious artifacts. The study was approved by the Human Research Ethics Committee of Khon Kaen University (HE654032).

B. Image acquisition and pre-processing

CT of KUB and abdomen was obtained by Siemens SOMATOM Definition DS (Dual Source 128) and CT Scanner or PHILIPS Brilliance iCT 128 CT Scanner on CT KUB and CT whole abdomen. To mimic real-world clinical practice, where kidney stone detection is primarily performed during the non-contrast phase, only these images were extracted from both CT KUB and whole abdomen scans. The original images were collected in a Digital Imaging and Communications in Medicine (DICOM) format in the PACS.

De-identified DICOM images were obtained from the medical records of patients diagnosed with kidney stones via CT scans performed between January 1, 2015, and December 31, 2020. Images were included only if they contained the

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kidneys within the field of view. A total of 1,642 images were extracted from 79 patients.

The DICOM images were then exported and converted to the JPEG format while maintaining de-identification. Subsequently, all images were resized to a uniform dimension of 640 x 640 pixels to ensure consistency during the deep learning model training process.

Two senior radiology residents and an attending radiologist independently categorized each image into two groups: "stone present" and "stone absent." All discrepancies were reviewed and resolved to ensure inter-rater agreement.

C. *Deep learning model development*

The images were divided into training, validation, and test sets using a stratified random sampling approach. This ensures a proportional representation of both "stone present" and "stone absent" categories within each set. Importantly, all images from the same patient were placed within a single set to prevent data leakage from the training set to the test set. The specific allocation is shown in Table 1.

TABLE I. DATASET SPLITTING FOR TRAINING, VALIDATION, AND TESTING

Training set: n=1330 (80%)	**Validation set: n=156 (10%)**	**Test set: n=156 (10%)**
Stone (+) : n=772	Stone (+) : n=83	Stone (+) : n=66
Stone (-) : n=558	Stone (-) : n=73	Stone (-) : n=90

We opted to utilize the original dataset size of 1,642 images to ensure a clear understanding of the model's performance on a representative sample of CT scans without the potential influence of artificially generated data. Data augmentation techniques were not employed

Kidney stone annotation on CT images was performed using Roboflow, a software platform for data annotation. Senior radiology residents utilized the polygon segmentation tool within Roboflow, which facilitated the precise delineation of kidney stone regions. All annotations were subsequently confirmed by an attending radiologist to ensure accuracy. Figure 1 depicts an example of a CT image with a segmented kidney stone.

For images devoid of kidney stones, a "null" label was assigned during annotation. Figure 2 illustrates an example of a CT image labeled as "null."

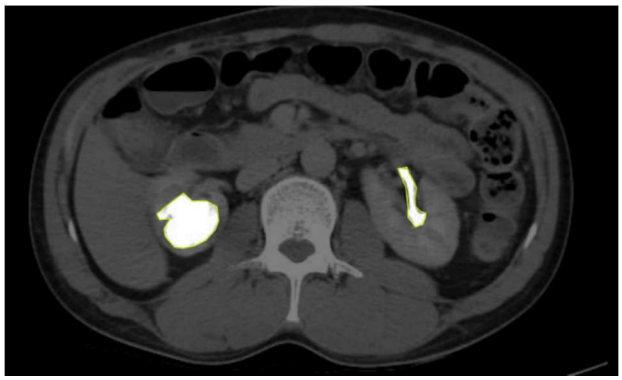

Fig. 1. Example of a CT Image with Polygon Segmented Kidney Stone.

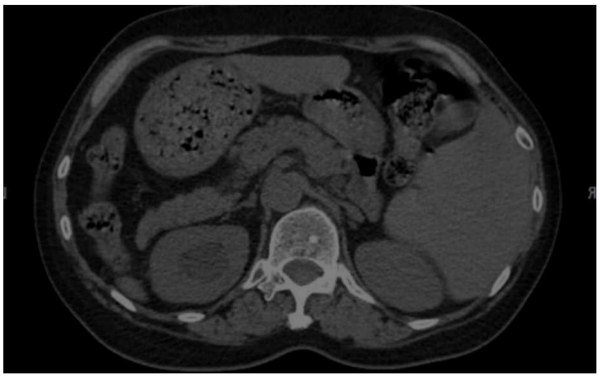

Fig. 2. Example of a CT Image Labeled as "Null".

Our study employed five pre-trained variants of the YOLOv8 object detection architecture: YOLOv8n (nano), YOLOv8s (small), YOLOv8m (medium), YOLOv8l (large), and YOLOv8x (extra). These models were selected for their ability to be fine-tuned for custom object detection tasks. We utilized Google Colab, a cloud-based platform, and various Python libraries to train each model on our kidney stone detection dataset.

The training process employed the following parameters:

Image Size: 640 x 640 pixels (ensuring consistency with image preprocessing)

Batch Size: 16 images (a common choice for balancing training speed and efficiency)

Optimizer: Stochastic Gradient Descent (SGD) – a widely used optimization algorithm for training deep learning models.

Epoch Selection: We employed a varying number of training epochs (iterations) for each model version based on their inherent complexity:

YOLOv8n and YOLOv8s: 50, 100, 150, 200, and 300 epochs

YOLOv8l: 50, 100, 150, and 200 epochs

YOLOv8x: 50 and 100 epochs

The rationale behind this approach is to provide sufficient training for smaller models while preventing overfitting for larger, more complex models.

The performance of each trained model was evaluated based on its diagnostic accuracy across the training, validation, and test sets. We employed primary performance evaluation metrics (which will be specified later in the report) to assess the effectiveness of each model in detecting kidney stones from CT scans.

The entire model training process was conducted on Google Colab, leveraging its cloud computing resources for efficient training, especially for larger models like YOLOv8l and YOLOv8x

III. RESULTS

A. *Patient characteristics*

A total of 69 patients were included in the study. The median age was 62 years (range 28-87 years). Forty-four patients (63.7%) were male.

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B. Model performance

The research results should be presented clearly and right to the point with accompanying figures and tables. These figures and tables should be referred to in the content. Explanation must not repeat what is already given in the content.

1) Performance of YOLOv8n Model: The YOLOv8n model was trained over epochs ranging from 50 to 300. The model achieved its best performance at the 150th epoch, as shown in Table 2. At this epoch, the model achieved an F1 score of 0.97, an F2 score of 0.97, a precision of 0.98, and a recall of 0.96. It also attained a mean Average Precision (mAP) of 0.992 at an Intersection over Union (IoU) threshold of 0.5, and 0.599 at an IoU threshold of 0.5:0.95. Additionally, the model demonstrated a high accuracy of 0.99.

TABLE II. DIAGNOSTIC PERFORMANCE OF THE YOLOV8N MODEL FOR DETECTED KUB STONE

Epoch	F1[1]	F2[2]	Precision[3]	Recall[4]	mAP[5] 50	mAP 50-95	Accuracy
50	0.94	0.94	0.941	0.933	0.97	0.606	0.95
100	0.95	0.95	0.924	0.968	0.971	0.587	0.97
150	0.97	0.97	0.980	0.96	0.992	0.599	0.99
200	0.94	0.94	0.959	0.925	0.968	0.575	0.93
300	0.93	0.93	0.932	0.920	0.968	0.566	0.93

[1]F1 score: A harmonic mean of precision and recall, balancing both metrics. [2]F2 score: Similar to F1 score, but weights recall more heavily. [3]Precision: The proportion of correctly identified kidney stones among all detections. [4]Recall: The proportion of true kidney stones correctly identified by the model. [5]Mean Average Precision (mAP): An overall measure of object detection accuracy across different IoU thresholds. IoU threshold: A parameter used to define how well a predicted bounding box overlaps with the ground truth bounding box.

2) Performance of YOLOv8s Model: Similar to the YOLOv8n model, the YOLOv8s model was trained over epochs rang-ing from 50 to 300. As shown in Table 3, the model achieved its optimal performance at the 200th epoch. At this epoch, the YOLOv8s model obtained an F1 score of 0.96, an F2 score of 0.96, a precision of 0.974, a recall of 0.933, a mean Average Precision (mAP) of 0.984 at an IoU threshold of 0.5, and 0.58 at an IoU threshold of 0.5:0.95. Additionally, the model ex-hibited an accuracy of 0.97.

TABLE III. DIAGNOSTIC PERFORMANCE OF THE YOLOV8S MODEL FOR DETECTED KUB STONE

Epoch	F1[1]	F2[2]	Precision[3]	Recall[4]	mAP[5] 50	mAP 50-95	Accuracy
50	0.93	0.93	0.910	0.948	0.971	0.587	0.92
100	0.94	0.94	0.983	0.907	0.972	0.587	0.93
150	0.92	0.92	0.962	0.880	0.964	0.580	0.92
200	0.96	0.96	0.974	0.933	0.984	0.580	0.97
300	0.93	0.93	0.985	0.891	0.963	0.568	0.92

[1]F1 score: A harmonic mean of precision and recall, balancing both metrics. [2]F2 score: Similar to F1 score, but weights recall more heavily. [3]Precision: The proportion of correctly identified kidney stones among all detections. [4]Recall: The proportion of true kidney stones correctly identified by the model.[5]Mean Average Precision (mAP): An overall measure of object detection accuracy across different IoU thresholds. IoU threshold: A parameter used to define how well a predicted bounding box overlaps with the ground truth bounding box.

3) Performance of YOLOv8m Model: Following the trend observed with smaller models, the YOLOv8m model achieved its best performance at an earlier epoch compared to larger models. As detailed in Table 4, the optimal performance was attained at the 50th epoch. During this epoch, the YOLOv8m model achieved an F1 score of 0.96, an F2 score of 0.96, a precision of 0.97, a recall of 0.96, a mean Average Precision (mAP) of 0.986 at an IoU threshold of 0.5 and 0.59 at an IoU threshold of 0.5:0.95, and an accuracy of 0.97.

TABLE IV. DIAGNOSTIC PERFORMANCE OF THE YOLOV8M MODEL FOR DETECTED KUB STONE

Epoch	F1[1]	F2[2]	Precision[3]	Recall[4]	mAP[5] 50	mAP 50-95	Accuracy
50	0.96	0.96	0.970	0.960	0.986	0.590	0.97
100	0.95	0.95	0.959	0.944	0.974	0.556	0.97
150	0.95	0.95	0.959	0.945	0.965	0.567	0.97
200	0.95	0.95	0.986	0.919	0.977	0.576	0.97
300	0.96	0.95	0.960	0.966	0.981	0.570	0.97

[1]F1 score: A harmonic mean of precision and recall, balancing both metrics. [2]F2 score: Similar to F1 score, but weights recall more heavily. [3]Precision: The proportion of correctly identified kidney stones among all detections. [4]Recall: The proportion of true kidney stones correctly identified by the model.[5]Mean Average Precision (mAP): An overall measure of object detection accuracy across different IoU thresholds. IoU threshold: A parameter used to define how well a predicted bounding box overlaps with the ground truth bounding box.

4) Performance of YOLOv8l Model: The YOLOv8l model exhibited its optimal performance at the 100th epoch, as shown in Table 5. This is consistent with the trend observed for larger models requiring fewer epochs for training compared to smaller models. During this epoch, the YOLOv8l model achieved an F1 score of 0.97, an F2 score of 0.97, a precision of 0.948, a recall of 0.987, a mean Average Precision (mAP) of 0.983 at an IoU threshold of 0.5 and 0.595 at an IoU threshold of 0.5:0.95, and an accuracy of 0.96.

TABLE V. DIAGNOSTIC PERFORMANCE OF THE YOLOV8L MODEL FOR DETECTED KUB STONE

Epoch	F1[1]	F2[2]	Precision[3]	Recall[4]	mAP[5] 50	mAP 50-95	Accuracy
50	0.95	0.95	0.968	0.933	0.970	0.585	0.95
100	0.97	0.97	0.948	0.987	0.983	0.595	0.96
150	0.94	0.94	0.958	0.920	0.977	0.570	0.93
200	0.96	0.96	0.947	0.973	0.983	0.568	0.97

[1]F1 score: A harmonic mean of precision and recall, balancing both metrics. [2]F2 score: Similar to F1 score, but weights recall more heavily. [3]Precision: The proportion of correctly identified kidney stones among all detections. [4]Recall: The proportion of true kidney stones correctly identified by the model.[5]Mean Average Precision (mAP): An overall measure of object detection accuracy across different IoU thresholds. IoU threshold: A parameter used to define how well a predicted bounding box overlaps with the ground truth bounding box.

5) Performance of YOLOv8x Model: Limited to 50 and 100 epochs due to its increased complexity, the YOLOv8x model achieved its best performance at the 100th epoch, as detailed in Table 6. During this epoch, the model exhibited exceptional results with an F1 score of 0.99, an F2 score of 0.99, a preci-sion of 0.987, a recall of 0.986, a mean Average Precision (mAP) of 0.99 at an IoU threshold of 0.5 and 0.6 at an IoU threshold of 0.5:0.95, and an accuracy of 0.99.

TABLE VI. DIAGNOSTIC PERFORMANCE OF THE YOLOV8X MODEL FOR DETECTED KUB STONE

Epoch	F1[1]	F2[2]	Precision[3]	Recall[4]	mAP[5] 50	mAP 50-95	Accuracy
50	0.97	0.97	0.970	0.973	0.986	0.611	0.97
100	0.99	0.99	0.987	0.986	0.990	0.600	0.99

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[1]F1 score: A harmonic mean of precision and recall, balancing both metrics. [2]F2 score: Similar to F1 score, but weights recall more heavily. [3]Precision: The proportion of correctly identified kidney stones among all detections. [4]Recall: The proportion of true kidney stones correctly identified by the model. [5]Mean Average Precision (mAP): An overall measure of object detection accuracy across different IoU thresholds. IoU threshold: A parameter used to define how well a predicted bounding box overlaps with the ground truth bounding box.

IV. Discussion

Our study investigated the performance of various YOLOv8 models for detecting kidney stones in CT images of the KUB and the whole abdomen. The results demonstrated promising potential, with all model variants (YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8x) achieving high diagnostic accuracy across multiple evaluation metrics, in-cluding F1 score, precision, recall, mean Average Precision (mAP), and accuracy.

Several prior studies have explored the application of deep learning for kidney stone detection in CT scans. Caglayan et al. (2022)(8) employed an xResNet50 deep learning model to detect kidney stones on unenhanced CT scans, achieving the most accuracy at 93%. Yildirim et al. (2021) (9) utilized a deep learning model and reported a 96.82% accuracy rate and 95.76% sensitivity on the 146 test cases. With more sophisticated techniques, using image processing techniques and geometry principles, Ebrahimi and Mariano created a semi-automated program to enhance kidney stone detection in computed tomography images(10). More recent studies have explored the application of radiomics in kidney stone detection tasks. For instance, a study by Perrot T(11) employed a machine learning classi-fication model trained on radiomics features to differentiate kidney stones from phlebo-liths in unenhanced low-dose computed tomography, achieving an accuracy of up to 85.1%.

Our study employed YOLOv8 models for kidney stone detection and achieved high diagnostic performance across all variants, with the YOLOv8x model attaining an excep-tional F1 score of 0.99. While direct comparisons between studies can be challenging due to variations in methodology and datasets, our findings demonstrate the potential of YOLOv8 models for accurate kidney stone detection, potentially offering comparable or even superior performance to existing AI models

The YOLOv8n model achieved its peak performance at the 150th epoch, exhibiting a robust balance between precision and recall (F1 score of 0.97) alongside high precision, re-call, mAP, and accuracy. Similarly, the YOLOv8s model displayed strong performance at the 200th epoch (F1 score of 0.96), effectively identifying and localizing kidney stones.

Interestingly, the YOLOv8m model reached its optimal performance earlier at the 50th epoch. Despite this, it maintained high F1 score, precision, recall, mAP, and accuracy values, demonstrating its efficacy in kidney stone detection. This observation suggests that for smaller model architectures, fewer training epochs might be sufficient to achieve opti-mal performance.

The YOLOv8l model also presented impressive diagnostic performance, with an F1 score of 0.97 at the 100th epoch. This finding highlights its capability for accurate kidney stone identification and localization.

Remarkably, the YOLOv8x model surpassed all other variants, achieving an excep-tional F1 score of 0.99 at the 100th epoch. This indicates a near-perfect balance between precision and recall, bolstered by high precision, recall, mAP, and accuracy values. The superior performance of the YOLOv8x model can be attributed to its increased complexity, allowing for more intricate feature extraction and improved object detection capabilities. However, it is important to acknowledge the trade-off between training time and perfor-mance, as the YOLOv8x model with its larger architecture was limited to only 50 and 100 epochs for training.

Our deep learning model exhibits several key strengths. Firstly, it demonstrates the ability to accurately detect kidney stones of varying sizes, including stones as small as 1 mm, through the use of boundary box detection, Figures 3-4. This capability highlights the model's potential for early and precise stone detection.

Secondly, the model effectively distinguishes between images containing kidney stones and those without stones, as demonstrated in Figures 3-5. This differentiation ca-pability is crucial for accurate diagnosis and efficient clinical workflow.

An additional strength of our study lies in the high quality of the data utilized to train and validate the model. All cases included in the dataset were confirmed surgical cases with pathological verification. Furthermore, the segmentation of kidney stones within the images was meticulously performed and confirmed by radiologists using polygon tools, ensuring the accuracy of stone pixel identification. This high-quality dataset likely con-tributed to the model's robust performance.

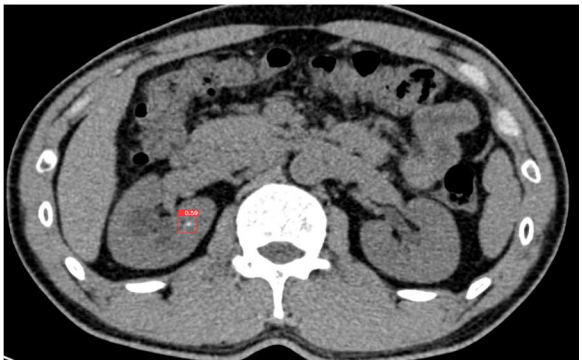

Fig. 3. Figure 3 illustrates the detection of a 1 mm renal stone, visualized by a bounding box, in the right kidney.

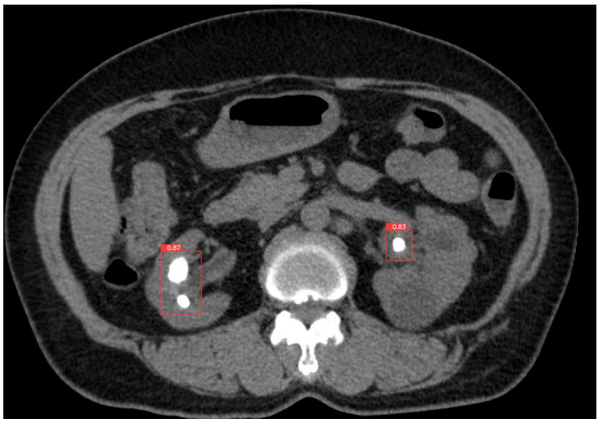

Fig. 4. shows a bounding box detection of stones in both kidneys.

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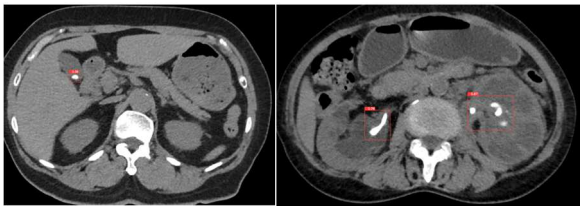
Fig. 5. shows no bounding box detection, signifying the absence of a kidney stone.

Our study identified limitations in the model's object detection capabilities for certain scenarios. Images containing interfering objects with hyperattenuating characteristics, such as gallstones (Figure 6a) or urinary catheters (Figure 6b), presented challenges. These objects could be misinterpreted as kidney stones by the model, potentially leading to false positive detections. However, these mistakes are readily rectified when combined with clinical judgment.



(**a**) (**b**)

Fig. 6. False Positive Detections: (a) Positive detection in an image containing a gallstone; (b) Case with a left renal stone and bilateral ureteral stents. The model generated a false positive detection in the region of the ureteral stents.

Some limitations require consideration for future research. First, our study did not account for the composition of kidney stones, which plays a crucial role in treatment deci-sions. Different stone types may exhibit varying responses to treatments, potentially im-pacting the AI algorithm's detection accuracy(12). As highlighted in the work by Fitri et al. (13), a CNN-based model achieved impressive accuracy (up to 0.991) in classifying uric acid, calcium, and mixed composition stones within the urinary tract system using micro-CT images. Future studies incorporating stone composition data can enhance the model's overall effectiveness and guide clinical decision-making. Second, we did not evaluate the AI algorithm's ability to predict treatment success for kidney stones. Under-standing this predictive power could offer valuable insights for clinicians. Further re-search incorporating treatment outcomes data can pave the way for the development of a comprehensive AI model that not only detects kidney stones but also predicts treatment success.

V. Conclusions

In conclusion, this study demonstrates the potential of YOLOv8 models for accurate kidney stone detection in CT scans. All models achieved high diagnostic performance, with the YOLOv8x model exhibiting exceptional results. However, limitations exist, in-cluding the lack of consideration for stone composition and treatment prediction. By ad-dressing these limitations in future research, we can refine the model to enhance its diag-nostic capabilities and predictive power, ultimately leading to improved clinical decision-making and patient outcomes.

Acknowledgment *(Heading 5)*

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